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Course Title IT645 Web and Mobile Development
Date September 2, 2026

Max Marks 25
Time 1 Hour

Instructions:

1. All questions are compulsory.
2. Write your answers briefly; writing long answers does not fetch higher marks.
3. Each question carries 5 marks.
4. Make suitable assumptions wherever necessary.

1. Consider the code below and answer the following questions.

```
<!DOCTYPE html>
<html>
<head>
<style>
.box {
  width: 150px;
  padding: 10px 20px;
  border: 5px solid black;
  margin: 10px;
  display: inline-block;
}
.box:hover {
  transform: scale(1.1);
}
</style>
</head>
<body>
<div class="box">Box 1</div>
<div class="box">Box 2</div>
</body>
</html>
```

- a. Calculate the total width occupied by each box, including padding, border, and margin. (2 marks)
- b. What is the effect of display: inline-block in this example?
- c. What happens when the mouse pointer is placed over a box?
- d. If box-sizing: border-box is added to .box, what will be the total width of the box excluding the margin?

2. Consider the following code

```
<button id="btn">Click Me</button>
<p id="result">0</p>
<script>
let count = 0;
document.getElementById("btn").addEventListener("click", function() {
  count++;
  document.getElementById("result").textContent = count;
  localStorage.setItem("count", count);
});
</script>
```

- a. What will be displayed in the paragraph after the button is clicked three times?
- b. What data is stored in localStorage after three clicks?
- c. Modify the code so that the value stored in localStorage is retrieved when the page is loaded.
- d. If sessionStorage is used instead of localStorage, what is the key difference in the lifetime of the stored data?

3. Consider the code below and answer

```

<meta name="viewport" content="width=device-
width, initial-scale=1.0">
<style>
.container {
  display: flex;
  gap: 20px;
}
.card {
  width: 300px;
  padding: 20px;
  background: lightblue;
}
@media (max-width: 600px) {
  .container {
    flex-direction: column;
  }
  .card {
    width: 100%;
  }
}
</style>

<div class="container">
  <div class="card">HTML</div>
  <div class="card">CSS</div>
  <div class="card">JavaScript</div>
</div>

```

- What is the purpose of the viewport `<meta>` tag?
- What will happen to the three cards when the browser width is 500px?
- What will happen when the browser width is 800px?
- Write code to make the cards appear in a row only when the screen width is at least 700px.

4. A college wants its course page to provide structured information about a course to search engines. The following JSON-LD code is incomplete. Answer the following

```

<script type="application/ld+json">
{
  "@context": "https://schema.org",
  "@type": "Course",
  "name": "Web Programming",
  "description": "Introduction to modern web
development",
  "provider": {
    "@type": "Organization",
    "name": "ABC University"
  }
}
</script>

```

- Add a suitable url property to the JSON-LD for the course page.
- Add a courseCode property with the value WP101.
- Why is JSON-LD useful even though it may not be directly visible on the webpage?
- Write a robots.txt file such that all search engines can crawl /courses/ but not /students/.

5. Consider the following HTML and CSS code. Study the code carefully and answer the questions that follow. Assume that the browser viewport is 1000 px wide and 800 px high:

```
<!DOCTYPE html>
<html>
<head>
<style>
.container {
  width: 80vw;
  height: 70vh;
  margin: 5vh auto;
  display: flex;
  flex-direction: row;
  justify-content: space-between;
  align-items: center;
  gap: 2em;
  position: relative;
  border: 2px solid black;
}
.box {
  height: 20vh;
  padding: 1em;
  border: 2px solid black;
  position: relative;
}
.box1 {
  flex-basis: 25%;
}
.box2 {
  flex-basis: 35%;
}
.box3 {
  flex-basis: 20%;
}
.badge {
  position: absolute;
  top: 1em;
  right: 1em;
  width: 5vw;
  height: 5vh;
}
</style>
</head>
<body>
<div class="container">
  <div class="box box1">Box 1</div>
  <div class="box box2">Box 2</div>
  <div class="box box3">Box 3</div>
  <div class="badge">Badge</div>
</div>
</body>
</html>
```

a. Draw the layout of the webpage showing the position of the three boxes and the badge. Clearly indicate the width and height of the container and the dimensions of each box.

b. Calculate the values in pixels for:

- 50vw
- 50vh
- 2em

c. Explain how flex-basis affects the initial width of .box1, .box2, and .box3.

d. The .badge has position: absolute. Relative to which element is its position calculated? Show its position in your diagram.

container
vh: 560
margin: 40
width: 800
gap: 2

box
vh: 160

badge
width: 50
height: 5

$$25 \times \frac{160}{100} = 40$$

$$20 \times \frac{160}{100} = 32$$

$$35 \times \frac{160}{100} = 56$$